FedPD: A Federated Learning Framework with Optimal Rates and Adaptivity to Non-IID Data

Xinwei Zhang⋆ Mingyi Hong⋆ Wotao Yin††
Sairaj Dhople⋆ Yang Liu†

⋆*University of Minnesota;*
† *Webank*
†† *UCLA*

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Federated Learning (FL)

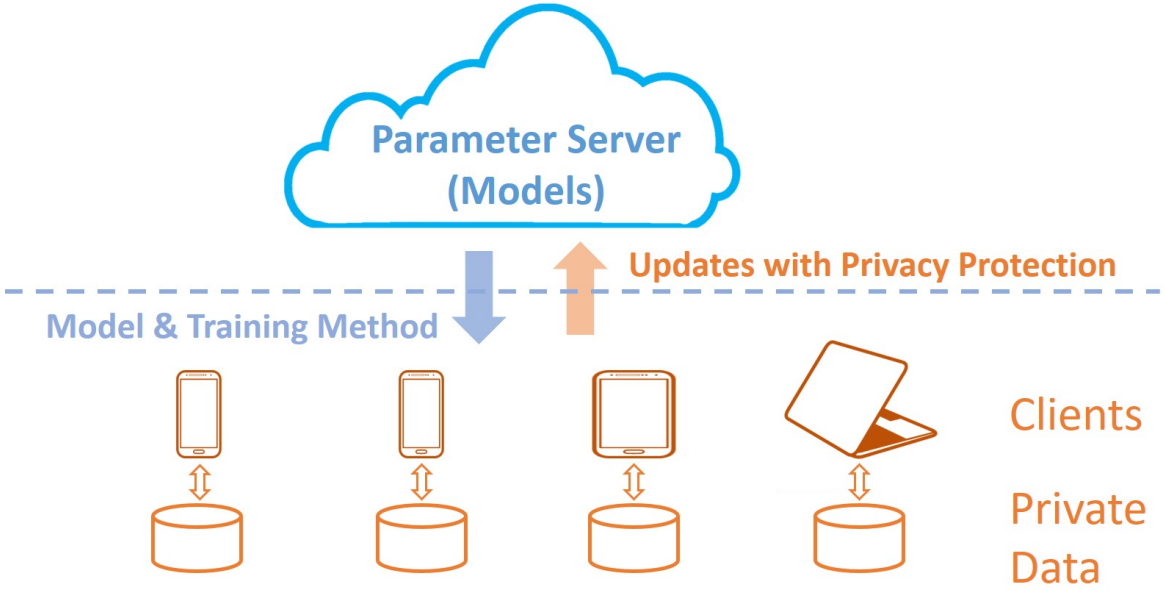


Figure 1: System structure of federated learning

- Parameter server network
- Massively distributed private data
- Limited Communication Capacity

System Structure

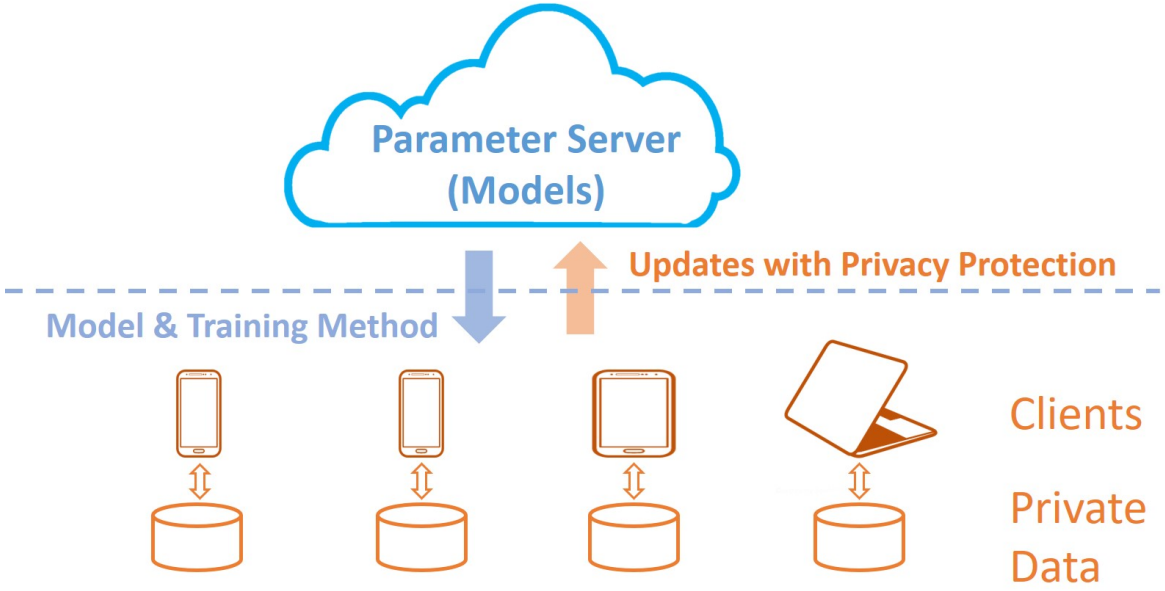
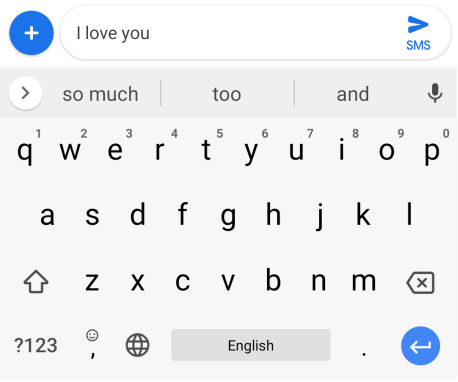


Figure 2: System structure of federated learning

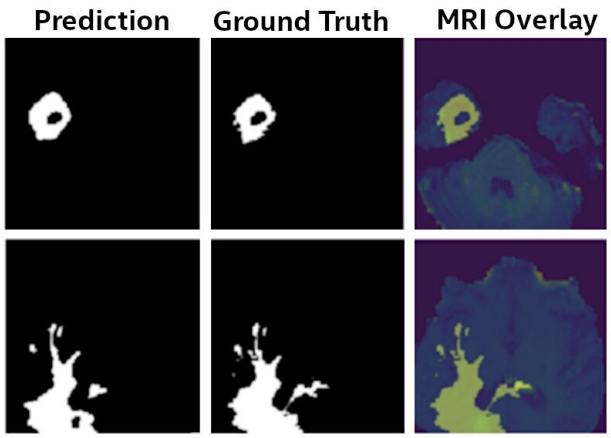
- Compute then Aggregate (CTA) protocol
  - Multiple local computation
  - Only exchange models; no gradient or other information exchange (better protect private data [ZMB20])

Applications

(a) Keyboard Prediction



(b) Medical Imaging



(c) Robot Control

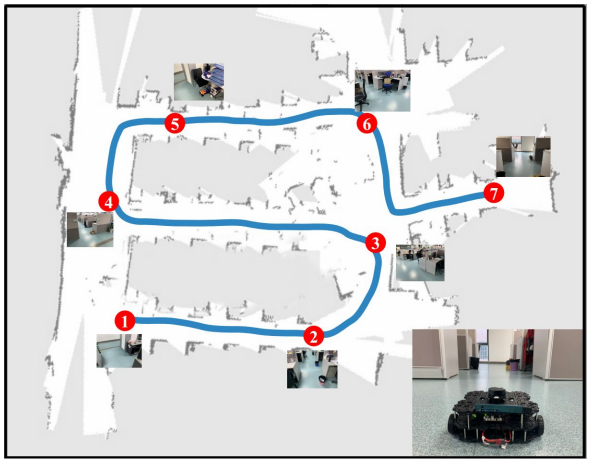


Figure 3: Applications of federated learning in keyboard prediction [Yan+18], medical imaging [She+19] and robot control [Liu+19]

Problem Formulation

The federated (model) learning problem is formulated as

$$\min_{\mathbf{x}\in\mathbb{R}^d} f(\mathbf{x}) := \frac{1}{N}\sum_{i=1}^{N} \underbrace{\frac{1}{|\mathcal{D}_i|}\sum_{\xi_i\in\mathcal{D}_i} F(\mathbf{x};\xi_i)}_{:=f_i(\mathbf{x})}, \tag{1}$$

$\mathbf{x}$: a d-dimensional model;
N: the number of agents;
$\mathcal{D}_i$: data set stored on the i-th agent;
ξ_i: one sample in $\mathcal{D}_i$;
$M := \sum_{i=1}^{N} |\mathcal{D}_i|$: total sample size;
$F : \mathbb{R}^d \to \mathbb{R}$ is the loss function for i-th data point;
Note: $f_i(\mathbf{x})$ can also be expressed as $f_i(\mathbf{x}) := \mathbb{E}_{\xi\in P_i} F(\mathbf{x};\xi)$

Problem Formulation

Typical questions that are of interests:

- What type of problems can we deal with (convex, non-convex? streaming? etc.)
- What's the aggregation protocol (model exchange only, or more info needed?)
- How flexible local updates are (full gradient updates? SGD? multiple rounds?)
- Solution quality (diverges? global min? etc.)
- How fast the algorithm converges (communication efficiency? sample complexity? etc.)

This Work

- Provide some theoretical understanding about the FL problem (from optimization perspective)
- Better understand the popular FedAvg algorithm
- Develop an algorithmic framework that addresses a few questions posed just now

General Assumptions

Assumption 1

(Smoothness)
Each $f_i(\cdot)$ in (1) is L–smooth.

$$\|\nabla f_i(\mathbf{x}) - \nabla f_i(\mathbf{y})\| \leq L \|\mathbf{x} - \mathbf{y}\|, \ \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^d, i \in [N]. \tag{2}$$

Clearly, this assumption leads to the following

$$\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \leq L \|\mathbf{x} - \mathbf{y}\|, \ \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^d. \tag{3}$$

Do not require convexity

General Assumptions

Assumption 2

(Lower-bounded)
The objective of problem (1) *satisfies:*

$$f(\mathbf{x}) \geq c > -\infty, \quad \text{for some constant } c, \forall\, \mathbf{x} \in \mathbb{R}^d. \tag{4}$$

Assumption 1-2 are common and very mild

Special Assumptions in FL Literature

Assumption 3

(Bounded Gradient (BG))
The gradients of the local functions are upper bounded as follows

$$\|\nabla f_i(\mathbf{x})\|^2 \leq G^2, \quad \forall\, \mathbf{x} \in \mathbb{R}^d, i = 1, \ldots, N, \tag{5}$$

where $G > 0$ is some fixed constant.

Strong assumption. Not true for typical learning problems

Example. Consider the simple linear regression problems (where $\mathbf{A}_i$ and $\mathbf{b}_i$ are related to data).

$$f_i(\mathbf{x}) = \|\mathbf{A}_i\mathbf{x} - \mathbf{b}_i\|^2$$

Special Assumptions in FL Literature

Assumption 4

(I.I.D. Data)

(a) - *The local functions are i.i.d. if the following holds:*

$$\nabla f_i(\mathbf{x}) = \nabla f_j(\mathbf{x}), \quad \forall\, \mathbf{x} \in \mathbb{R}^d,\ \forall\, i \neq j, \tag{6}$$

(b) - *The following holds (for some fixed $B > 0$)*

$$\sum_{i=1}^{N} \|\nabla f_i(\mathbf{x})\|^2 \leq B^2 \|\nabla f(\mathbf{x})\|^2, \forall\, \mathbf{x},\ \text{s.t. } \|\nabla f(\mathbf{x})\|^2 > \epsilon. \tag{7}$$

Again, strong assumption!

In FL, data (such as medical records, keyboards input data) is naturally generated by the individual agents.

Special Assumptions in FL Literature

A reasonable relaxation of i.i.d. is the following:

Assumption 5

(δ-Non-I.I.D. Data) *The local functions are δ -non-i.i.d. if the following holds:*

$$\left\|\nabla f_i(\mathbf{x}) - \nabla f_j(\mathbf{x})\right\| \leq \delta, \quad \forall\, \mathbf{x} \in \mathbb{R}^d,\ \forall\, i \neq j. \tag{8}$$

By varying δ from 0 to ∞, it provides a useful characterization of data non-i.i.d.-ness.

FedAvg Algorithm

Algorithm 1 FedAvg Algorithm [Kon+16]

Input: Total iteration # T, initial point $\mathbf{x}^0$, local iteration # Q.
Initialize: $\mathbf{x}_i^0 \triangleq \mathbf{x}^0, i = 1, \ldots, N$
for $r = 0, \ldots, T-1$ (global iterations) **do**
  for $q = 0, \ldots, Q-1$ (local iterations) **do**
    for $i = 1, \ldots, N$ in parallel **do**
      either **Option 1** (Local SGD)
      or **Option 2** (Local GD)
    end for
  end for
  Global averaging: $\mathbf{x}_0^{r+1} = \frac{1}{N}\sum_{i=1}^N \mathbf{x}_i^{r,Q}$
  $\mathbf{x}_i^{r+1,0} = \mathbf{x}_0^{r+1}, i \in [N]$
end for
Output: Randomly samples $\mathbf{x}^r \in \{\mathbf{x}_0^0, \ldots, \mathbf{x}_0^T\}$.

FedAvg Algorithm

Two typical choices of the local update steps:

Option 1 (Local SGD):

Sample $\xi_i^{r,q}$ form $\mathcal{D}_i$,

Set $\mathbf{x}_i^{r,q+1} = \mathbf{x}_i^{r,q} - \eta^{r,q} \nabla F(\mathbf{x}_i^{r,q}; \xi_i^{r,q})$

Option 2 (Local GD):

$$\mathbf{x}_i^{r,q+1} = \mathbf{x}_i^{r,q} - \eta^{r,q} \nabla f_i(\mathbf{x}_i^{r,q}).$$

Open Question 1

Q1 (local directions).

What are the most effective local update directions?

- FedAvg: multiple local GD (SGD)
  $$\mathbf{x}_i^{r,q+1} = \mathbf{x}_i^{r,q} - \eta^{r,q} \underbrace{\nabla f_i(\mathbf{x}_i^{r,q})}_{\text{direction}}$$

- FedProx [Sah+18]: multiple perturbed local GD (SGD)
  $$\mathbf{x}_i^{r,q+1} = \mathbf{x}_i^{r,q} - \eta^{r,q} \underbrace{(\nabla f_i(\mathbf{x}_i^{r,q}) + \mu(\mathbf{x}_i^{r,q} - \mathbf{x}_0^r))}_{\text{direction}}$$

Open Question 2

Q2 (global aggregation).

Does sophisticated aggregation help improve performance?

- FedAvg: simple averaging
  $\mathbf{x}_0^{r+1} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{x}_i^{r,Q}$
- Agnostic FedAvg [MSS19]: weighted averaging
  $\mathbf{x}_0^{r+1} = \frac{1}{\sum_{i=1}^{N} w_i}\sum_{i=1}^{N} w_i \mathbf{x}_i^{r,Q}$

Simple averaging most popular; Not clear if other types of (linear) processing is helpful.

Open Question 3

Q3 (communication efficiency).

When and how do multiple local updates reduce the communication overhead?

To achieve some $\mathcal{O}(\epsilon)$- optimal solution

- FedAvg [YJY19]: $O(1/\epsilon^{3/2})$ aggregation steps with $O(1/\epsilon^{1/2})$ local GD steps in between;
- FedAvg [WJ18]: $O(1/\epsilon)$ aggregation steps with $O(1/\epsilon)$ local SGD steps in between
- VRL-SGD [Lia+19]: $O(1/\epsilon)$ aggregation steps with $O(1/\epsilon)$ local SGD steps in between

Open Question 4

Q4 (assumptions).

Best performance of FL algorithms with minimum assumptions?

- FedAvg: Either BG or I.I.D-(a) or both;
- FedProx: I.I.D-(b);
- VRL-SGD: Bounded sample gradient variance == BG
- ...

Investigate Open Problems

Communication, computation and aggregation

- Fundamental tradeoff between communication and local computation
- For FL using CTA protocol, does more local computation help?
- What's the best possible communication complexity (i.e., rounds of model exchanges)?
- A generic FL algorithm model has to be introduced

Communication, computation and aggregation

- Optimality condition (ϵ-stationarity)

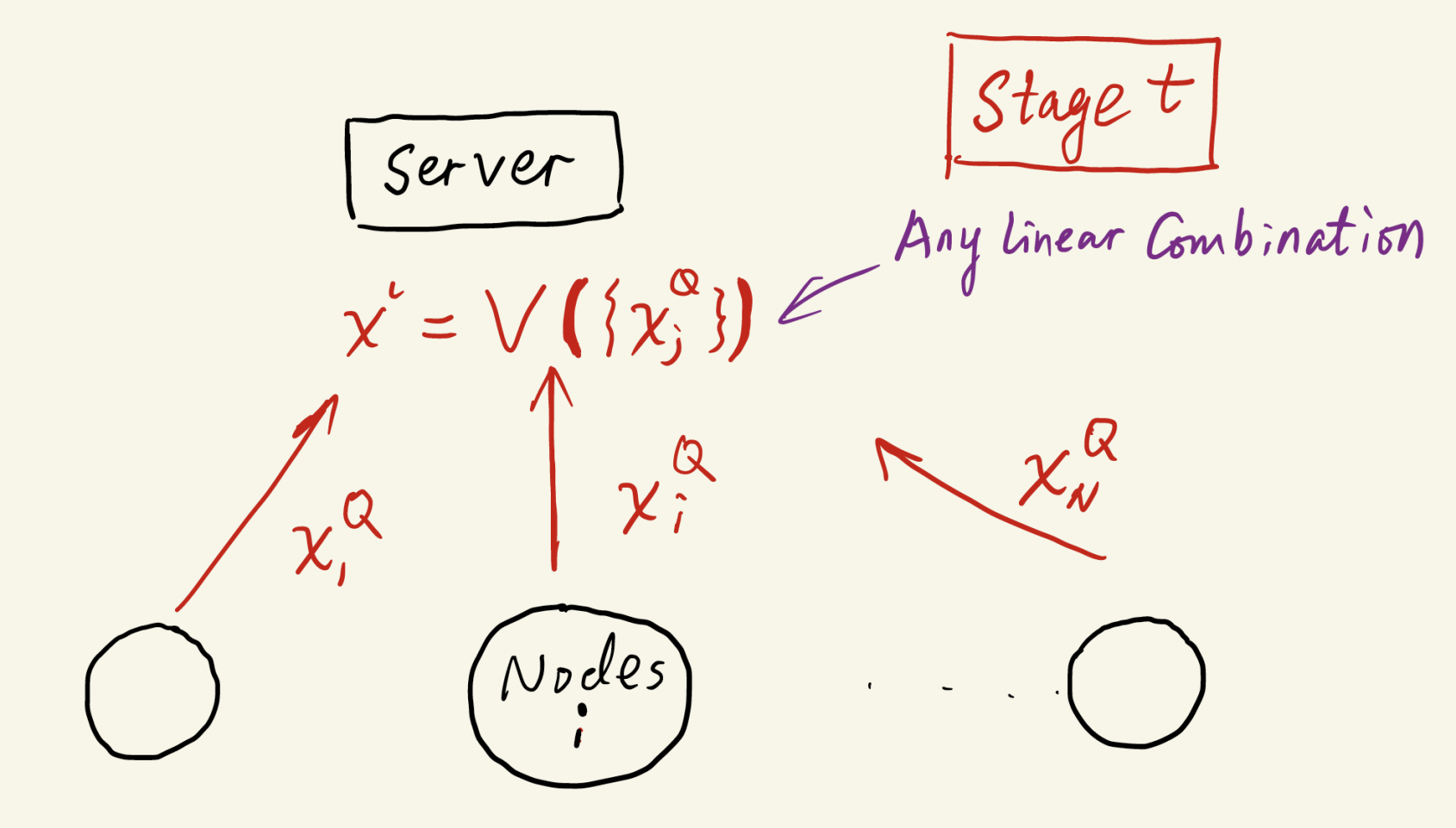
$$\|\nabla f(\mathbf{x}^*)\|^2 \leq \epsilon$$

- A Generic CTA model:

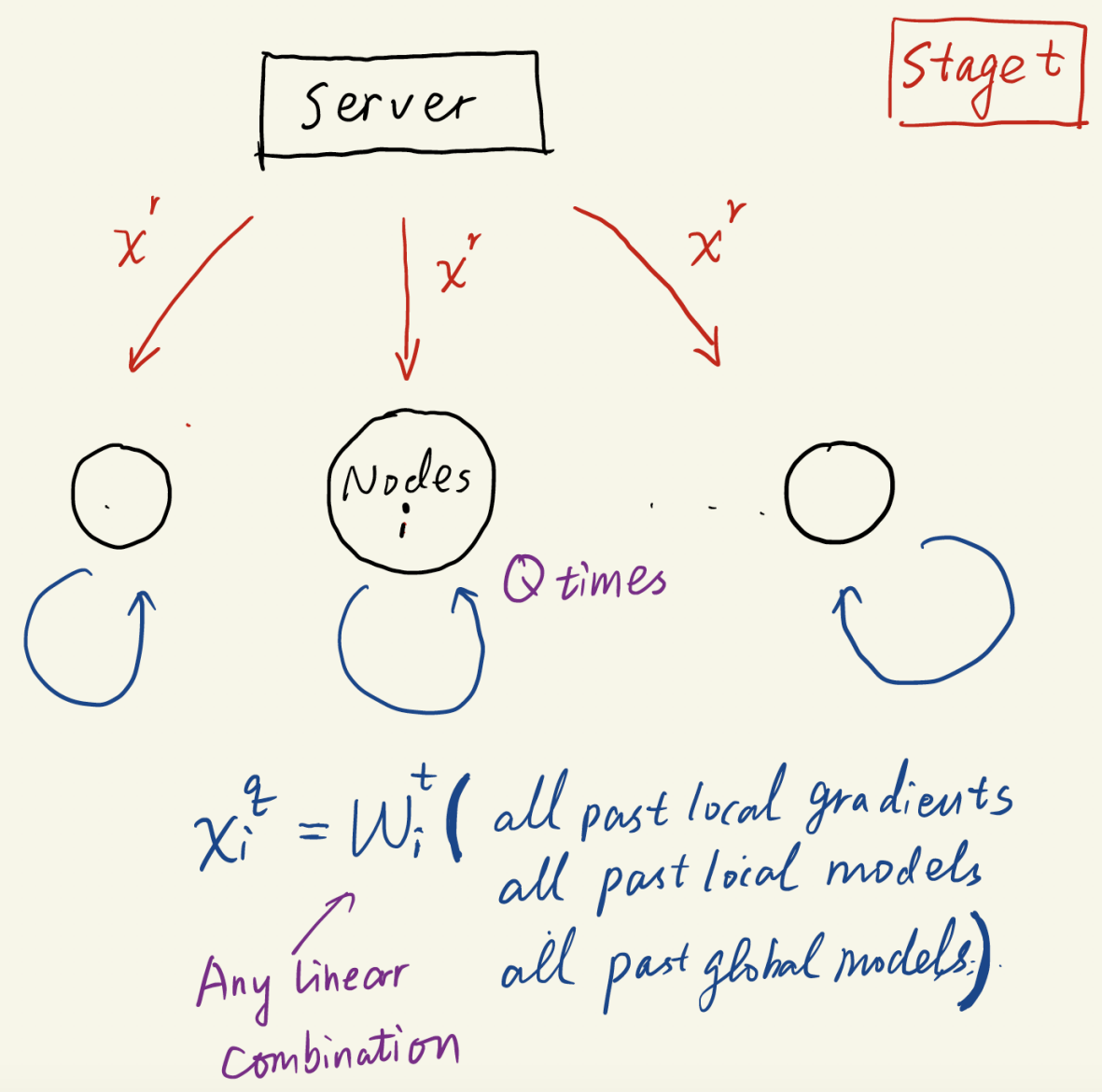
$$x^r = V^r(\{x_i^{r-1,Q}\}_{i=1}^N),\ x_i^{r,0} = x^r,\ \ \forall\, i \in [N] \tag{9}$$

$$x_i^{r,q} \in W_i^r\left(\{x_i^{\tau,k}, \left\{\nabla F(x_i^{\tau,k};\xi_i)\right\}_{\xi_i \in D_i}\}_{k\in[q-1],\tau\in[r]}\right),$$
$$\forall\, q \in [Q],\ \forall\, i \in [N]. \tag{10}$$

The Generic CTA Model



The Generic CTA Model



A Lower Bound

Claim 3.1

(Informal) *There exists a problem instance satisfying Assumptions 1 – 2, such that for any $Q > 0$, any algorithm in "CTA"-class takes at least $\mathcal{O}(1/\epsilon)$ communication rounds to reach ϵ-stationary solution.*

- Better (linear) processing does not help
- Local computation does not help
- $\mathcal{O}(1/\epsilon)$ communication complexity the best one can do

Best Local Directions and Assumptions

- Is performing local (stochastic) GD always the best choice?
- Does the i.i.d., bounded gradient assumption really necessary?
- Investigate the popular FedAvg algorithm

Divergence of FedAvg

Claim 3.2

Suppose that Assumption 1-2 holds true, but without BG, or without both BG and i.i.d. Then FedAvg with local-GD and local SGD can diverge to infinity for any $Q > 1$.

- Both BG and i.i.d. are essential for FedAvg
- Meaningless solution will be generateds
- Why this happens? Centralized algorithm will not have this; Because dad directions? or we should not perform averaging?

Algorithm Designing

Federated Primal-Dual (FedPD) Algorithm

- Algorithm that address Q1-Q4
  - Achieve the best communication complexity ($\mathcal{O}(1/\epsilon)$)
  - Achieve the best computation complexity
  - Require only Assumption 1-2
- Adaptivity & Flexibility
  - Flexible local solver (GD/SGD/etc.)
  - Communication reduction for δ-non-i.i.d data

Problem Reformulation

An equivalent reformulation:

$$\min_{\mathbf{x}_0, \mathbf{x}_i} \frac{1}{N} \sum_{i=1}^{N} f_i(\mathbf{x}_i), \text{ s.t. } \mathbf{x}_i = \mathbf{x}_0, \ \forall i = 1, \ldots, N. \tag{11}$$

$\mathbf{x}_0$ at server, $\mathbf{x}_i$'s at the agents;

Similar to ADMM [Boy+11], define augmented Lagrangian (AL)

$$\mathcal{L}(\mathbf{x}_0, \mathbf{x}_1, \ldots, \mathbf{x}_N, \lambda) \triangleq \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}_i(\mathbf{x}_0, \mathbf{x}_i, \lambda_i),$$

$$\mathcal{L}_i(\mathbf{x}_i, \mathbf{x}_0, \lambda_i) \triangleq f_i(\mathbf{x}_i) + \underbrace{\langle \lambda_i, \mathbf{x}_i - \mathbf{x}_0 \rangle}_{\text{dual variable}} + \underbrace{\frac{1}{2\eta} \|\mathbf{x}_i - \mathbf{x}_0\|^2}_{\text{penalty}}.$$

Federated Primal-Dual Algorithm

Algorithm 2 Federated Primal-Dual (FedPD) Algorithm

Input: $\mathbf{x}^0, \eta, R, T, Q_1, \ldots, Q_N$, Initialize: $\mathbf{x}_0^0 = \mathbf{x}^0$,
for $r = 0, \ldots, T-1$ **do**
  for $i = 1, \ldots, N$ in parallel **do**
    Local Update:
    $\mathbf{x}_i^{r+1} = \text{Oracle}_i(\mathcal{L}_i(\mathbf{x}_i^r, \mathbf{x}_{0,i}^r, \lambda_i^r), Q_i)$ [flexible local update]
    $\lambda_i^{r+1} = \lambda_i^r + \frac{1}{\eta}(\mathbf{x}_i^{r+1} - \mathbf{x}_{0,i}^r)$, $\mathbf{x}_{0,i}^{r+} = \mathbf{x}_i^{r+1} + \eta\lambda_i^{r+1}$
  end for
  if $(r+1 \mod R) = 0$ **then**
    Aggregate: $\mathbf{x}_0^{r+1} = \frac{1}{N}\sum_{i=1}^N \mathbf{x}_{0,i}^{r+}$ [simple averaging]
    Broadcast: $\mathbf{x}_{0,i}^{r+1} = \mathbf{x}_0^{r+1}$
  else
    Local Update: $\mathbf{x}_{0,i}^{r+1} = \mathbf{x}_{0,i}^{r+}$ [skip comm.]
  end if
end for

Federated Primal-Dual Algorithm

- Flexible in the choice of $\text{Oracle}_i(\cdot)$
- Oracles defines algorithmic properties
  1. Can perform either *multiple* or a *single* local updates
  2. Communication frequency R adaptive to the degree of non-i.i.d-ness
  3. Can reduce the total sample complexity (SC)
- Remove BG and I.I.D assumption
- Real communication rounds T/R

Oracle I (Local GD/SGD)

Algorithm 3 Oracle Choice I

Input: $\mathcal{L}_i(\mathbf{x}_i^r, \mathbf{x}_{0,i}^r, \lambda_i^r), Q_i$
Initialize: $\mathbf{x}_{i,0}^r = \mathbf{x}_i^r$,
Option I (GD):
  for $q = 0, \ldots, Q_i - 1$ **do**
    $\mathbf{x}_i^{r,q+1} = \mathbf{x}_i^{r,q} - \eta_1 \nabla_{\mathbf{x}_i} \mathcal{L}(\mathbf{x}_i^{r,q}, \mathbf{x}_{0,i}^r, \lambda_i^r)$
  end for
Option II (SGD):
  for $q = 0, \ldots, Q_i - 1$ **do**
    $\mathbf{x}_i^{r,q+1} = \mathbf{x}_i^{r,q} - \eta_1 \left(h_i(\mathbf{x}_i^{r,q}; \xi_i^{r,q}) + \lambda_i^r + \frac{1}{\eta}(\mathbf{x}_i^{r,q} - \mathbf{x}_{0,i}^r) \right)$
  end for
Output: $\mathbf{x}_i^{r+1} \triangleq \mathbf{x}_i^{r,Q_i}$

Q_i's are large enough so that $\left\| \nabla_{\mathbf{x}_i} \mathcal{L}(\mathbf{x}_i^{r,Q}, \mathbf{x}_{0,i}^r, \lambda_i^r) \right\|^2 \leq \epsilon$.

Main Results - Convergence

Theorem 1

Suppose Assumptions 1 –2 hold, and consider FedPD with Oracle I. Suppose $0 < \eta < \frac{\sqrt{5}-1}{4L}$*, and Assumption 5 holds with some finite* $\delta > 0$ *then we have:*

$$\frac{1}{T}\sum_{r=0}^{T}\|\nabla f(\mathbf{x}_0^r)\|^2 \leq \mathcal{O}(1/T) + \mathcal{O}\left(\frac{(C_3^{(R-1)}-1)^2}{R}\right)\delta^2 + \mathcal{O}(\epsilon), \qquad (12)$$

Otherwise, if Assumption 5 holds with $\delta = \infty$*, when* $R = 1$*, we have*

$$\frac{1}{T}\sum_{r=0}^{T}\|\nabla f(\mathbf{x}_0^r)\|^2 \leq \mathcal{O}(1/T) + \mathcal{O}(\epsilon),$$

$C_3 > 1$ *is a constant related to* L *and independent of* T, δ, R*.*

Main Results - Communication

- To reach ϵ accuracy, $T = \mathcal{O}(1/\epsilon)$ stages
- When the data is completely non-i.i.d. ($\delta = \infty$), then $R = 1$, communication complexity $T/R = \mathcal{O}(1/\epsilon)$.
- When δ is small that $R \approx \log(\frac{\epsilon}{\delta^2}) > 1$, then communication complexity is

$$T/R = \mathcal{O}\left(\frac{1}{(\epsilon \log(\frac{\epsilon}{\delta^2}))}\right).$$

Intuitively makes sense – i.i.d. data, no communication needed!
Completely non-i.i.d. data, no savings possible!

Oracle II (Variance Reduction (VR))

Algorithm 4 Oracle Choice II

Input: $\mathcal{L}_i(\mathbf{x}_i^r, \mathbf{x}_{0,i}^r, \lambda_i^r), Q, I, B$

Initialize: $\mathbf{x}_i^{r,0} = \mathbf{x}_i^r$,

if $r \mod I = 0$ **then**

$g_i^{r,0} = \nabla f_i(\mathbf{x}_i^{r,0})$

else

$g_i^{r,0} = g_i^{r-1,Q}$

end if

for $q = 0, \ldots, Q-1$ **do**

$\mathbf{x}_i^{r,q+1} = \arg\min_{\mathbf{x}_i} \tilde{\mathcal{L}}_i(\mathbf{x}_i, \mathbf{x}_{0,i}^r, \lambda_i^r; \mathbf{x}_i^{r,q}, g_i^{r,q})$

$g_i^{r,q+1} = g_i^{r,q} + \frac{1}{B}\sum_{b=1}^{B}(h_i(\mathbf{x}_i^{r,q+1}; \xi_{i,b}^{r,q}) - h_i(\mathbf{x}_i^{r,q}; \xi_{i,b}^{r,q}))$

end for

Output: $\mathbf{x}_i^{r+1} = \mathbf{x}_i^{r,Q}, v_i^{r,Q}$

Oracle II (Variance Reduction (VR))

$\tilde{\mathcal{L}}_i^r(\mathbf{x}_i)$ is the linearized AL function at iteration $\mathbf{x}_i^{r,q}$, and the optimizer has an explicit expression:

$$\mathbf{x}_i^{r,q+1} = \frac{\eta}{\eta+\gamma}\mathbf{x}_i^{r,q} + \frac{\gamma}{\eta+\gamma}\mathbf{x}_{0,i}^r - \frac{\eta\gamma}{\eta+\gamma}(g_i^{r,q} + \lambda_i^r),$$

where γ is a constant.

I: Frequency to compute full gradient
B: Minibatch size in evaluating stochastic gradient
Total SC:

$$M + M * T/I \text{ (Full gradient) } + 2BN * QT \text{ (Stochasitc gradient)}.$$

Note: This oracle is *not* a variation of Oracle I.

Main Results - Convergence

Theorem 2

Suppose Assumptions 1 –2 hold, and consider FedPD with Oracle II. Choose $R = 1$, and η satisfies $\eta \in \left(0, \frac{1}{3(Q+\sqrt{QI/B})L}\right)$, $\gamma > \frac{5\eta}{B\sqrt{L}}$. Then the following holds:

$$\frac{1}{T}\sum_{r=0}^{T}\mathbb{E}\left\|\nabla f(\mathbf{x}_0^r)\right\|^2 \leq \mathcal{O}(1/T). \tag{13}$$

Main Results - Computation

- To reach ϵ accuracy, $T = \mathcal{O}(1/\epsilon)$
- The total SC: $M + M * T/I + 2BN * QT$
- $I = \sqrt{M}$, $B = \sqrt{M}/(QN) \rightarrow$

$$SC = \mathcal{O}\left(M + \frac{\sqrt{M}}{\epsilon}\right)$$

- Compare with
  - FedAvg, VRL-SGD: $SC = \mathcal{O}(\frac{1}{\epsilon^2})$
  - FedProx: $\mathcal{O}(\frac{M}{\epsilon})$
  - F-SVRG: $SC = \mathcal{O}(\frac{\sqrt{M}}{\epsilon})$ with gradient communication.

Numerical Results

- FEMNIST data set (naturally non-i.i.d. data)
- Train neural network model

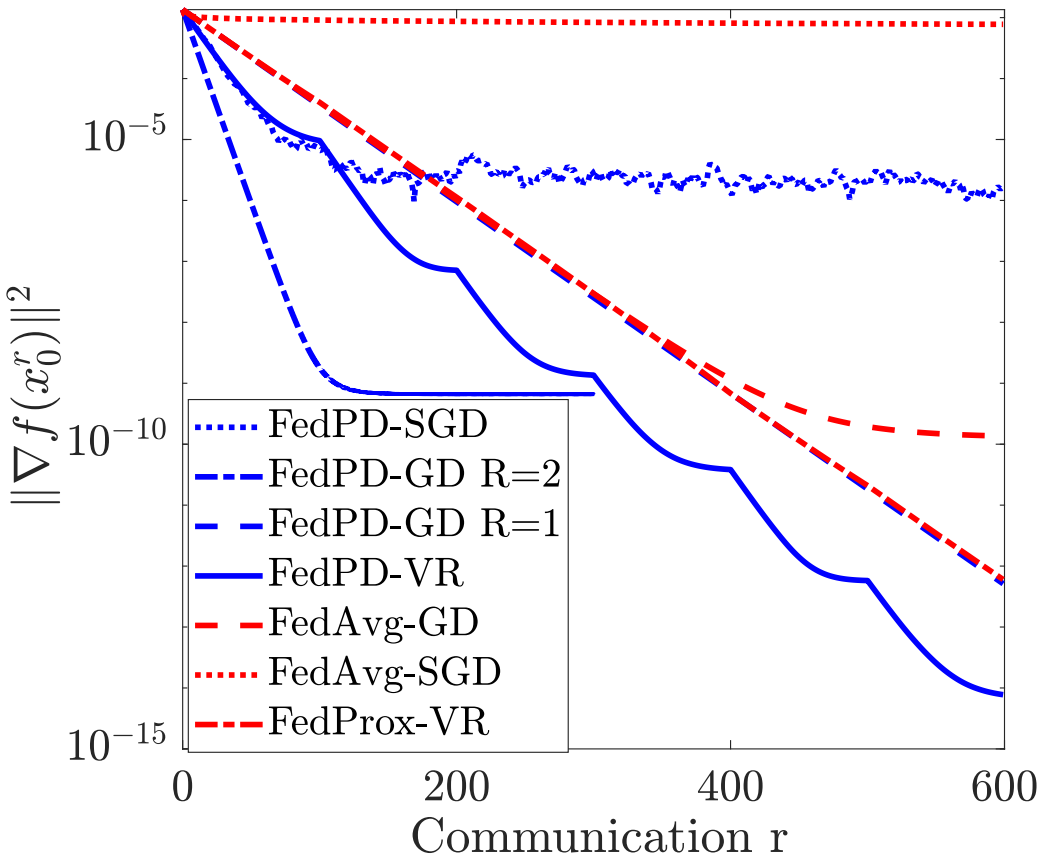


Figure 4: Convergence of FedPD, FedAvg and FedProx w.r.t. iteration.

Numerical Results

- FEMNIST data set (naturally non-i.i.d. data)
- Train neural network model

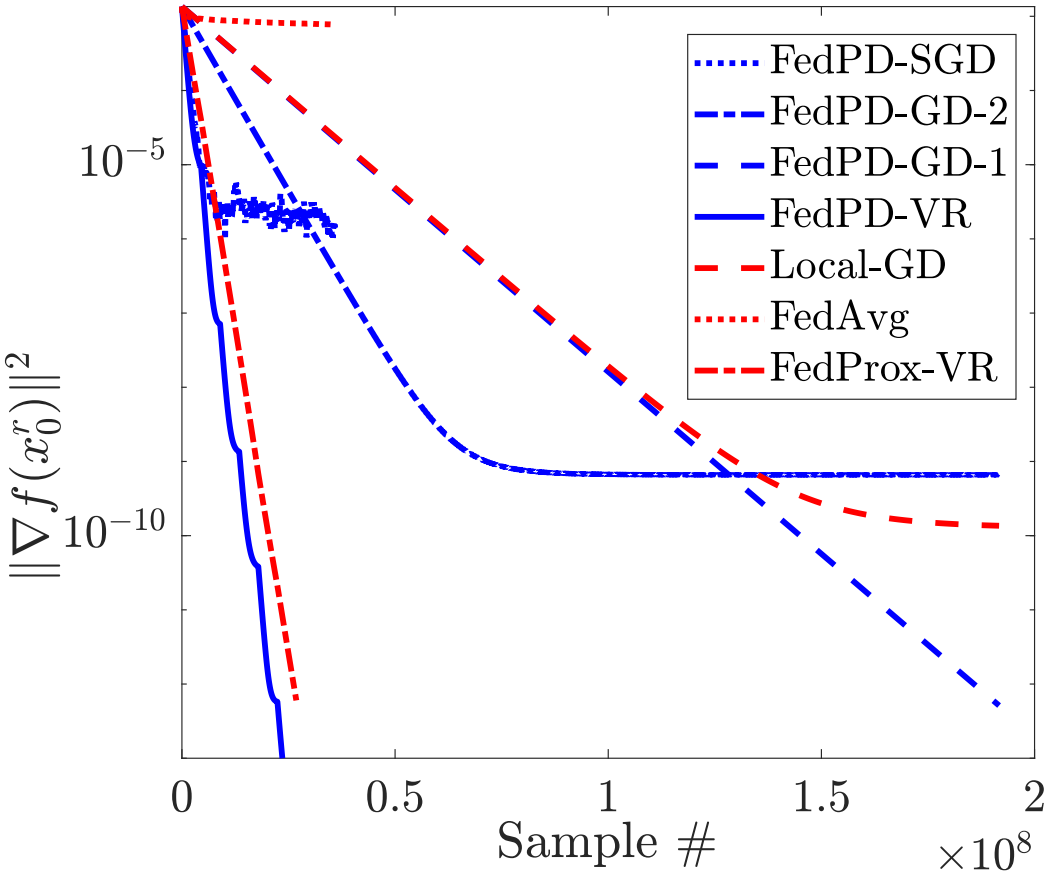


Figure 5: Convergence of FedPD, FedAvg and FedProx w.r.t. sample.

Summary

FedPD as a FL meta-algorithm.

- Flexible local solvers (deterministic/stochastic)
- Adaptive to system requirements
- Reduce communication according to data distribution
- Reduce computation complexity

Future Works

Other type of comm. scheme, sparsification, quantization

- Heterogeneity makes things challenging
- Recent work [Chen et al 19] shows adding noise to Sign based transmission reduces heterogeneity

Privacy and security issues

Incorporating learning based criteria into the lower bound characterization?

Summary

Algorithm	Conv.	DN	BG	comm. (T)	comp. (QT)	sample
FedAvg	μSC	0	No	$\mathcal{O}\left(\frac{1}{\epsilon^{1/2}}\right)$	$\mathcal{O}(1/\epsilon)$	$\mathcal{O}(1/\epsilon)$
FedAvg	μSC	-	Yes	$\mathcal{O}\left(\frac{1}{\epsilon}\right)$	$\mathcal{O}(1/\epsilon)$	$\mathcal{O}(1/\epsilon)$
Coop-SGD	NC	0	No	$\mathcal{O}(1/\epsilon)$	$\mathcal{O}(1/\epsilon^2)$	$\mathcal{O}(1/\epsilon^2)$
MPR-SGD	NC	-	Yes	$\mathcal{O}(1/\epsilon^{3/2})$	$\mathcal{O}(1/\epsilon^2)$	$\mathcal{O}(1/\epsilon^2)$
Local-GD	C	-	No	$\mathcal{O}(1/\epsilon^{3/2})$	$\mathcal{O}(1/\epsilon^2)$	$\mathcal{O}(M/\epsilon^2)$
FedProx	NC	-$\star$	No	$\mathcal{O}(1/\epsilon)$	$\mathcal{O}(1/\epsilon^2)$	$\mathcal{O}(1/\epsilon^2)$
F-SVRG	NC	-	No	$\mathcal{O}(1/\epsilon)$	$\mathcal{O}(Q/\epsilon)$	$\mathcal{O}(M/\epsilon)$
VRL-SGD	NC	-	Yes	$\mathcal{O}(1/\epsilon)$	$\mathcal{O}(1/\epsilon^2)$	$\mathcal{O}(1/\epsilon^2)$
FedPD-GD	NC	δ	No	$\mathcal{O}\left(\frac{1}{\epsilon c(\delta)}\right)$	$\mathcal{O}\left(\frac{\log(1/\epsilon)}{\epsilon}\right)$	$\mathcal{O}(\frac{M}{\epsilon})$
FedPD-SGD	NC	δ	No	$\mathcal{O}(\frac{1}{\epsilon c(\delta)})$	$\mathcal{O}(1/\epsilon^2)$	$\mathcal{O}(1/\epsilon^2)$
FedPD-VR	NC	-	No	$\mathcal{O}(1/\epsilon)$	$\mathcal{O}(Q/\epsilon)$	$\mathcal{O}(\sqrt{M}/\epsilon)$

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